

[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 6 August 2026, 10:25 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:33 AM**Time taken** 7 mins 41 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int i=1,s=1,count=2;//initializing i and s
6      while(s<=n){
7          count++;//for each while loop
8          i++;
9          s+=i;
10         count+=2;//increment i and adding s
11     }
12     count++;//end of while loop
13     printf("%d",count);
14     return 0;
15 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 6 August 2026, 10:34 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:33 AM**Time taken** 58 mins 24 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int count = 0; // Global counter variable
4
5  void func(int n) {
6      count++; // For the 'if(n==1)' condition check
7      if (n == 1) {
8          count++; // For printf("*")
9      } else {
10         // Outer loop
11         for (int i = 1; i <= n; i++) {
12             count++; // For outer loop condition check
13
14             // Inner loop
15             for (int j = 1; j <= n; j++) {
16                 count++; // For inner loop condition check
17                 count++; // For first printf("*")
18                 count++; // For second printf("*")
19                 count++; // For break statement
20                 break;
21             }
22         }
23         count++; // For the final outer loop condition check (when i becomes n+1 and fails)
24     }
25 }
26
27 int main() {
28     int n;
29     scanf("%d",&n);
30
31     func(n);
32
33     printf("%d", count);
34     return 0;
35 }
```

36 |

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 6 August 2026, 11:38 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:40 AM**Time taken** 2 mins 45 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int count=0;
3  void Factor(int num)
4  {
5      for (int i = 1; i <= num; ++i)
6      {
7          count++; //i<=num
8          count++; //if condition
9          if (num % i == 0)
10         {
11             count++; //printf("%d ", i);
12         }
13     }
14     count++; //for loop failed
15 }
16 int main(){
17     int num;
18     scanf("%d", &num);
19     Factor(num);
20     printf("%d", count);
21     return 0;
22 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Thursday, 6 August 2026, 11:41 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:47 AM**Time taken** 6 mins 36 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int count=0;
3  void function(int n)
4  {
5      int c = 0;
6      count++;
7      for(int i=n/2; i<n; i++){
8          count++;//condition check
9          for(int j=1; j<n; j = 2 * j)
10         {
11             count++;//condition check
12             for(int k=1; k<n; k = k * 2)
13             {
14                 count++;//condition check
15                 c++;
16                 count++;//increment
17             }
18             count++;//condition failed for k loop
19         }
20         count++;//condition failed for j loop
21     }
22     count++;//condition failed for i loop
23 }
24 }
25 int main(){
26     int n;
27     scanf("%d",&n);
28     function(n);
29     printf("%d",count);
30     return 0;
31 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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[Harini S B 251801088-2025-AIDS](#) ▾**H2****Started on** Monday, 24 August 2026, 1:07 PM**State** Finished**Completed on** Monday, 24 August 2026, 1:18 PM**Time taken** 11 mins 45 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int reverse(int n)
3  {
4      int rev = 0, remainder;
5      int count=1;
6      while (n != 0)
7      {
8          count++;
9          remainder = n % 10;
10         count++;
11         rev = rev * 10 + remainder;
12         count++;
13         n/= 10;
14         count++;
15     }
16     count++;
17     count++;//printf(rev);
18     return count;
19 }
20 int main(){
21     int n;
22     scanf("%d",&n);
23     printf("%d",reverse(n));
24 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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